AZIZ KAPDAWALA

Senior Data Engineer | Cloud Data Platforms · ETL & Warehousing · Applied GenAI

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SUMMARY

Data engineer, close to four years across FinTech and U.S. healthcare. Currently own an AWS log-analytics and applied-AI platform end to end: Terraform-managed infrastructure, a dual-index OpenSearch model separating analytics aggregations from vector retrieval, and a Bedrock agent that root-causes automation failures for under a cent per run. Before that, Oracle-to-PySpark and ADF/Databricks warehouse migrations for a mutual fund and one of India's largest private banks.

EXPERIENCE

AppXcelerate Solutions Pvt Ltd — Nov 2025 – Present

Senior Software Engineer | Bengaluru

AI-integrated monitoring, alerting and root-cause analysis for a U.S. healthcare payer's RPA estate

- Took the RPA observability programme from a blank page to production after a database-level integration with the incumbent platform stalled — set the roadmap, wrote the delivery stories, shipped every phase since.
- Architected the platform on AWS — API Gateway webhook, SQS, Lambda, OpenSearch, Bedrock, SNS, DynamoDB, all in Terraform — giving operations one searchable view of 87 automated processes, 56 transaction queues and 48 robots across 21 account tenants.
- Built the ingestion path for at-least-once delivery under failure — HMAC-verified webhook, dead-letter queues, idempotent writes keyed on event identity, index templates and retention as code — and modeled two indices against two access patterns: keyword/numeric for the aggregation-heavy dashboards and SLA monitors, vector with Titan embeddings for semantic retrieval.
- Designed and shipped the root-cause engine — a Bedrock (Claude Haiku 4.5) agent that assembles evidence from aggregation queries, validates every citation against the source document before it reaches the answer, retrieves comparable past incidents from a vector memory, and posts a structured card into Slack behind a one-click button. Roughly $0.009 per analysis against an agreed ceiling, behind SSM flags so any part rolls back without a deploy.
- Restructured the repository into reusable Terraform modules, one root per environment behind a CI quality gate, with GitHub OIDC replacing long-lived AWS keys. When an organisation-wide policy denied the managed search service, re-platformed onto a hardened self-managed node in a private VPC with no internet-facing endpoint — keeping the programme on schedule.
- Python and query engineering throughout: boto3 and pandas in the Lambda services, OpenSearch DSL aggregations behind every metric reported, SQL against source systems for reconciliation, a Slack slash command for plain-English questions, and the Power BI model behind the monthly automation review.

Axis Mutual Fund — Jul 2024 – Aug 2025

Data Engineer | Mumbai

- Rebuilt the KFintech reconciliation for SEBI compliance as an automated validation pipeline, authoring 100+ advanced SQL validation rules that replaced the RTA operations team's manual checking entirely and gave compliance a reproducible audit trail.
- Migrated legacy Oracle stored-procedure logic to PySpark ETL on AWS EMR, cutting processing time 60% — through partitioning strategy, broadcast joins and replacing row-by-row cursor logic, not added cluster capacity.
- Prototyped the firm's first generative-AI use case on Bedrock and SageMaker, ran the Java Spring Boot services on Linux that several reporting flows depended on, and reviewed the juniors' pull requests.

Celebal Technologies — Mar 2023 – Jun 2024

Associate Data Engineer | Pune

- Delivered the source-to-lake migration for one of India's largest private banks, bringing Oracle, DB2 and Hadoop into a single Azure data platform.
- Built ingestion in Azure Data Factory to ADLS and the transformation layer in Databricks (PySpark, Spark SQL), modeling curated Delta tables that cut downstream processing time roughly 80% against the jobs they replaced.
- Led decommissioning of a Hitachi Pentaho estate that had hit a hard on-premises scaling ceiling — the work that got the bank's nightly batch finishing inside its window again.

TECHNICAL SKILLS

Data Warehousing & Modeling: Dimensional modeling (star/snowflake schemas), slowly changing dimensions, partitioning and clustering, incremental and CDC loads, query tuning and execution-plan analysis, Snowflake, Amazon Redshift, Databricks SQL, Delta Lake

SQL: Oracle PL/SQL, MS SQL Server T-SQL, PostgreSQL, MySQL, DB2, stored procedures, window functions

Python & ETL/ELT: PySpark, pandas, NumPy, boto3, pytest; Apache Spark, Azure Data Factory, AWS EMR, Airflow, idempotent and dead-letter ingestion, schema and retention as code

AI / GenAI: Amazon Bedrock (Claude, Titan), RAG, embeddings and vector search, agentic tool-use pipelines, prompt design, citation/grounding validation, offline eval harnesses, LLM cost control

Cloud & Platform: AWS (S3, Lambda, EMR, Redshift, SQS, SNS, API Gateway, OpenSearch, DynamoDB, SSM, CloudWatch, IAM, KMS), Azure (ADF, ADLS, Databricks), Terraform, GitHub Actions + OIDC, Linux, Git

BI: Power BI (DAX, data modeling, custom themes), Qlik Sense · **Also:** Java, Spring Boot, shell scripting, Jira

EDUCATION & CERTIFICATIONS

Vellore Institute of Technology — *Master of Computer Applications (MCA)* — **Aug 2024 – Aug 2026**

DG Ruparel College, Mumbai University — *B.Sc. Computer Science, CGPA 7.71* — **Jul 2019 – Apr 2022**

Certifications: HackerRank SQL (Advanced) · HackerRank SQL (Intermediate) · Complete SQL Mastery · Machine Learning A–Z (Python & R)